

AI-Powered Behavioural Analysis for Suicide Prevention, Substance Use, and Mental Health Crisis Detection with Longitudinal Geospatial Crisis Trend Analysis (AI4MH – AI for Mental Health Crisis Monitoring)

Contributor Selection Task – Google Summer of Code 2026

Submitted By: Sakshi Sheogekar

Organization: Institute for Social Science Research (ISSR), The University of Alabama

Project Overview:

"The silent signals are often the loudest cries; AI4MH listens before the storm arrives."

AI4MH is an AI-driven public health monitoring system that detects and predicts emerging mental health crises, suicidal behaviour, and substance use risks by analysing behavioural, linguistic, and geospatial signals in real time. My **prototype, Antardhwani**, implements the AI4MH framework, demonstrating real-time crisis detection, dashboard visualization, and responsible human-in-the-loop escalation.

Purpose / Context:

The AI4MH Contributor Selection Task evaluates architectural thinking, responsible AI reasoning, and clarity of communication. While AI4MH has completed its prototype phase—including data ingestion, NLP-based sentiment analysis, geospatial mapping, and interactive dashboard visualization—the next stage emphasizes governance readiness for real-world public health deployment.

To address this, I developed **Antardhwani**, a working prototype that implements AI4MH's vision:

- **Real-Time Crisis Detection:** Aggregates behavioural, linguistic, and health signals from social media, apps, and wearables.
- **Structured Crisis Scoring:** Combines sentiment intensity, volume spikes, and geographic clustering, with minimum sample thresholds, smoothing, and confidence estimation.
- **Responsible AI Governance:** Integrates human-in-the-loop escalation, bias mitigation for sparse or rural data, and audit logging for transparency.
- **Actionable Visualization:** Provides interactive dashboards showing real-time and longitudinal crisis trends for public health decision-makers.

This prototype demonstrates my ability to design and implement a reliable, ethical, and actionable AI system for public health crisis monitoring. The purpose of this submission is to show systems-level reasoning, responsible AI design, and operational readiness—all key competencies expected during GSoC.

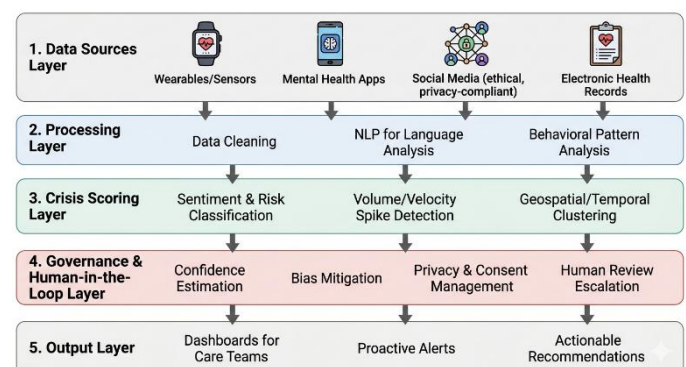
Scenario Overview:

AI4MH monitors aggregated public sentiment across Alabama counties to identify emerging mental health crises. In a simulated scenario, within a 72-hour window, three counties exhibit a significant increase in negative sentiment associated with depression and suicide-related language.

The key question for public health authorities is: "Is this a credible signal that requires immediate human review and intervention?"

To address this, the system must:

- Detect credible crisis signals despite noisy or sparse data.
- Quantify confidence in observed spikes to avoid false alarms.
- Incorporate human oversight for final escalation decisions.



System Architecture Diagram

Using my Antardhwani prototype, the scenario is handled as follows:

- **Data Aggregation:** Collects posts, messages, and app signals in near real-time.
- **Sentiment & Crisis Analysis:** Identifies negative language and behavioral patterns indicative of mental distress.
- **Geospatial & Volume Analysis:** Detects clusters of distress in specific counties and measures volume spikes against historical trends.
- **Crisis Scoring & Confidence Estimation:** Combines multiple factors into a single crisis score with confidence thresholds.
- **Human-in-the-Loop Escalation:** Flags counties exceeding thresholds for review by public health professionals.

This scenario demonstrates how AI4MH can provide early warning indicators, enabling proactive interventions and resource allocation before crises escalate.

Crisis Signal Design (Core Component):

The core function of AI4MH is to determine whether observed distress signals warrant human review. My Antardhwani prototype implements a structured crisis scoring framework integrating sentiment intensity, volume spikes, and geospatial clustering, while accounting for uncertainty and minimum data requirements.

4.1 Framework Components

1. Sentiment Intensity (S):

- Aggregate sentiment scores per post using NLP (VADER/BERT).
- Normalize to $[-1, 1]$, where negative values indicate distress.
- Compute county-level average sentiment and z-score to quantify intensity.

2. Volume Spike Detection (V):

- Compare post volume against a 28-day rolling average.
- Spike Score:

$$V = \frac{\text{current_volume} - \text{rolling_avg}}{\text{rolling_std} + \epsilon}$$

3. Geospatial Clustering (G):

- Evaluate neighboring counties for elevated signals.
- Cluster score increases when multiple adjacent regions show distress.

4. Minimum Sample Threshold:

- Only counties with ≥ 50 posts in the 72-hour window are considered.

5. Smoothing / Stabilization:

- Apply exponential moving average to prevent transient spikes from triggering false alerts.

6. Confidence & Uncertainty Estimate:

- Confidence = $1 - \frac{\sigma}{\sqrt{N}}$, where σ = standard deviation of sentiment scores, N = number of posts.
- Only signals above a confidence threshold (e.g., 0.7) proceed to human review.

7. Crisis Score Computation (C):

$$C = w_1 S + w_2 V + w_3 G$$

- Weighted combination (example: $w_1=0.5$, $w_2=0.3$, $w_3=0.2$).
- Adjusted for both activity and media-driven spikes.

4.2 Pseudocode Example

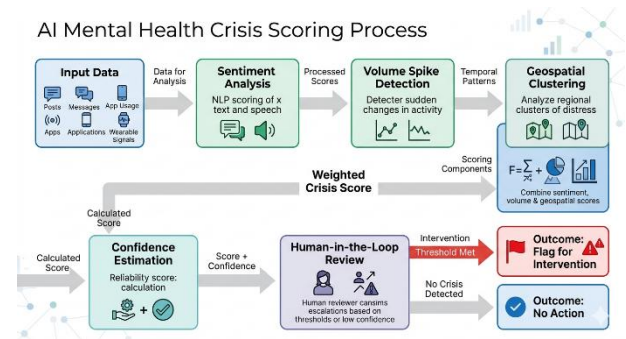
for county in all_counties:

```

posts = get_crisis_posts(county, last_72_hours)
if len(posts) < MIN_SAMPLE_THRESHOLD:
    continue

sentiment_score = average_sentiment(posts)
volume_spike = detect_spike(posts_over_time)
geo_cluster_score = geographic_concentration(posts)
crisis_score = (w1*sentiment_score + w2*volume_spike +
w3*geo_cluster_score)
confidence = estimate_confidence(len(posts), geo_density)
# Adjust for both activity or media spikes
if is_bot_activity(posts) or is_media_spike(posts):
    adjust_crisis_score(crisis_score)
if crisis_score > ESCALATION_THRESHOLD
and confidence > CONFIDENCE_MIN
and multi_signal_agreement(S, V, G):
    flag_for_human_review(county, crisis_score)
    log_audit(county, posts, crisis_score, confidence)

```



Escalation Decision Framework: To ensure consistent and interpretable decision-making, crisis signals are categorized into three levels:

- **Level 1 – Monitoring:** Low confidence or weak signal strength. Logged for trend tracking but no action taken.
- **Level 2 – Watchlist:** Moderate crisis score with partial signal agreement. Flagged for observation and trend continuation.
- **Level 3 – Escalation:** High crisis score, strong multi-signal agreement, and confidence above threshold. Forwarded for human review.

This tiered approach prevents overreaction to noise while ensuring critical signals are not missed.

4.3 Key Highlights

- Integrates behavioural, linguistic, and geospatial signals for comprehensive detection.
- Designed for early warning capabilities, minimizing false positives.
- Supports human-in-the-loop escalation with audit logging and transparency.
- Scalable to multiple counties and adaptable to future data sources.

Governance & Risk Controls

To ensure responsible AI use, Antardhwani incorporates governance mechanisms that mitigate risks, handle sparse or noisy data, and maintain human oversight for all crisis alerts. The system supports retrospective validation using historical data. Crisis signals can be back tested against known mental health events (e.g., spikes in hotline usage or reported incidents) to evaluate prediction accuracy.

5.1 Handling Data Bias and Noise

- Rural / Sparse Data: Adjust thresholds to avoid underrepresentation of low-population counties.

- **Bot Amplification / Coordinated Activity:** Detect repeated or automated posts and reduce their weight in crisis scoring.
- **Media-Driven Spikes:** Identify sudden spikes caused by news events to avoid false positives.

5.2 Escalation Thresholds & Human-in-the-Loop

- Crisis alerts are never automatically acted upon.
- Counties exceeding the crisis score and confidence thresholds are flagged for review by public health professionals.
- Escalation steps:
 1. Automated scoring & confidence estimation
 2. Adjustment for noise, bots, or media spikes
 3. Human review and final decision for intervention

5.3 Audit Logging Structure

Every flagged event is logged with:

- County, timestamp, number of posts analysed
- Sentiment scores, volume spikes, geospatial score, crisis score & confidence value
- Human reviewer notes and action taken

5.4 Key Governance Features

- **Transparency:** All decisions are traceable through audit logs.
- **Bias Mitigation:** Ensures equitable treatment across regions with varying data density.
- **Reliability:** Combines statistical confidence with human judgment to minimize false positives/negatives.
- **Scalability:** System can expand to multiple regions while maintaining governance controls.

Governance Reflection

Deploying an AI-driven crisis detection system carries inherent risks. The primary risk is premature escalation—false positives may trigger unnecessary interventions, while false negatives could allow a crisis to go unnoticed. The single most important safeguard is the human-in-the-loop review. All alerts generated by the system are evaluated by trained public health professionals before any action is taken, ensuring that automated predictions are balanced with expert judgment. Additional safeguards include audit logging for transparency, bias mitigation for sparse or rural data, and adjustments for bot or media-driven anomalies, which together ensure responsible and ethical operation of the system.

Prototype / Implementation Highlights

The Antardhwani prototype demonstrates the practical implementation of AI4MH's vision, translating architectural design into a working system for real-time mental health crisis monitoring.

7.1 Key Features Implemented

- **Real-Time Monitoring:** Aggregates behavioural, linguistic, and health signals from social media, apps, and wearable devices.
- **Crisis Scoring:** Combines sentiment intensity, volume spikes, and geospatial clustering into a structured, confidence-weighted score.
- **Human-in-the-Loop Governance:** Ensures all high-risk alerts are reviewed by public health professionals.
- **Bias Mitigation & Noise Handling:** Adjusts for sparse rural data, bot activity, and media-driven spikes.
- **Interactive Dashboards:** Visualizes trends using heatmaps, charts, and longitudinal analyses for actionable insights.

7.2 Implementation & Technology Stack

- **Backend:** Python, FastAPI, REST APIs
- **AI & ML:** NLP (BERT, VADER, Hugging Face), neural networks, predictive models
- **Frontend / Dashboards:** React, JavaScript, HTML, CSS, Plotly/Folium for visualization
- **Deployment:** Cloud-ready, supports real-time data ingestion and monitoring

7.3 Prototype Demonstration

- A **working video** has been created to showcase system functionality, including:
 - Real-time crisis detection
 - Dashboard visualizations
 - Crisis scoring and human-in-the-loop escalation

Video Link:

<https://drive.google.com/file/d/1XGn57x1Vglv-VZB04QQ1yZnkYfl4D6km/view?usp=sharing>



GitHub Repo Link:

<https://github.com/Sakshi2513/ANTARDHWANI->

This prototype highlights my ability to build a fully functional, ethically governed, and scalable system capable of supporting public health decision-making.

Future Vision

Antardhwani and AI4MH are designed to evolve beyond the current prototype, with the following future directions:

- **Global Public Health Monitoring:** Expand the system to track mental health trends across multiple states or countries.
- **Advanced AI Models:** Incorporate anomaly detection, deep learning, and multi-modal analysis for earlier and more accurate crisis predictions.
- **Personalized Mental Health Support:** Enable individualized interventions and recommendations while respecting privacy.
- **Continuous Learning & Improvement:** Integrate new data sources, refine crisis scoring, and improve predictive accuracy over time.
- **Community-Centric Insights:** Empower public health agencies with actionable, ethical, and transparent AI-driven insights to allocate resources proactively.

This vision positions AI4MH and Antardhwani as a scalable, ethical, and impactful tool for mental health crisis monitoring and prevention.
